DATASCI 406 FA 26, Quiz 3.4

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Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

Each Flint home gives a difference D_i, first draw minus flushed. The sign-flip test uses H_0: each D_i is symmetric about 0. What does the test condition on, and what does it rearrange?

1. Conditions on $|D_i|$; flips the signs, all patterns equally likely.
2. Conditions on $|D_i|$; makes each sign + with the observed fraction of positive D_i.
3. Conditions on the first-draw values; shuffles the flushed values among homes.
4. Conditions on nothing; resamples the D_i with replacement.

Solution

Answer: 1. Under symmetry about 0, D_i and $-D_i$ are equally likely, so given $|D_i|$ the sign is a fair coin, independently for each home. Swapping a home's two bottles flips its sign. Option 2: H_0 makes each sign a fair coin, and the observed fraction builds the alternative into the null. Option 3 breaks the pairing, which would test whether the two bottles are related, not whether flushing matters. Option 4 is the bootstrap, which does not impose H_0.

Question 2 (1 point)

A two-sample permutation test of $H_0 : F = G$ uses the difference in means. A student keeps the same label shuffles and computes the difference in medians instead. Which is true of the new test?

1. Under $F = G$ it still rejects at most 5% of the time; its power can change.
2. Under $F = G$ it no longer rejects 5% of the time, since the median has no simple null distribution.
3. It gives the same p-value, since the shuffles are the same.
4. It tests equal medians, so it rejects 5% of the time whenever the medians are equal.

Solution

Answer: 1. The shuffles come from H_0, and they control how often a true H_0 is rejected, whatever the statistic. The statistic decides which differences between F and G the test is likely to detect. Option 2: no formula for the null is needed, since the shuffles give it. Option 3: different statistics on the same shuffles give different p-values, as with the means, medians and KS in Part D.2. Option 4: the null is still $F = G$.

Question 3 (1 point)

Before computers, books printed tables of the rank-sum test's null distribution, but not of the permutation null of the difference in means. Why could the rank-sum null be tabulated?

1. Ranks are approximately normal, so one normal table serves every sample.
2. Without ties, its null depends only on m and n, not on the data.
3. The rank sum is always symmetric about 0, so half the table suffices.
4. The difference in means needs σ^2, which a table cannot know.

Solution

Answer: 2. Under $F = G$ the ranks of the first sample are a random subset of size m of $\{1, \ldots, N\}$, whatever the data are. The null of the difference in means depends on the pooled values. Option 1: the normal approximation is a large-sample result, not why exact tables exist. Option 3: the rank sum is centered at $m(N + 1)/2$, not 0. Option 4: the permutation null needs no σ^2; it depends on all the pooled values.

Question 4 (1 point)

In Part D.3, casual riders on the 231 other days had an SD of about 870, against 390 on the 500 working days. After centering each group at 0, the permutation test of the difference in means rejected 14% of the time at 5%, and Welch's statistic about 6%. Why does the difference in means reject too often?

1. $B = 400$ shuffles is too few, so the p-values are too small.
2. Casual riders are skewed, so means are less stable than medians.
3. Shuffling pools the two spreads, so the permutation SE is too small when the smaller group has the larger spread.
4. Centering each group made the means unequal, so H_0 was false.

Solution

Answer: 3. The groups have equal means but $F \neq G$, so the labels are not exchangeable. The permutation variance of $\bar{X} - \bar{Y}$ uses the pooled spread, $\hat{\sigma}^2_{\text{pooled}}(1/m + 1/n)$, while the actual variance is $\sigma_X^2/m + \sigma_Y^2/n$, which is larger here because the large spread sits in the small group. Welch's statistic uses each group's own variance. Option 1: B adds Monte Carlo error, not bias. Option 2: skew does not explain 14%. Option 4: centering made the means equal.

Rubric. 1 pt per question, no partial credit.